

AMAN PARGANIHA

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SUMMARY

AI/ML Engineer (M.Tech CSE, AI/ML) specializing in applied LLM systems, Retrieval-Augmented Generation (RAG), and multimodal document AI. Built an event-driven, containerized RAG pipeline and a credit-risk system curating ~41M SEC XBRL records into a 35,000-company dataset at 97.9% accuracy; merged open-source PRs in Matplotlib and JupyterLab.

EDUCATION

International Institute of Information Technology Naya Raipur India
Master of Technology in Computer Science & Engineering (AI/ML) | CGPA: 8.0/10.0 Aug 2025 – Aug 2027 (expected)
• Coursework: Generative AI, Deep Learning, Computer Vision, Data Science & Visualization, Cloud Computing

Shri Shankaracharya Technical Campus Bhilai India
Bachelor of Technology in Computer Science & Engineering | CGPA: 8.22/10.0 Aug 2019 – Aug 2023
• Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformer architectures, multimodal models, NLP, fine-tuning, model evaluation & benchmarking
LLMs & Generative AI: RAG pipelines, LlamaIndex, LangChain, OpenAI API (GPT-4o / 4o-mini), prompt engineering, embedding strategies (text-embedding-3-large), vector databases (Qdrant), agentic workflows, OCR / document AI
MLOps & Backend: Docker, AWS (EC2), FastAPI, REST APIs, event-driven orchestration (Inngest), asynchronous processing, Streamlit
Data Engineering & Analysis: Pandas, NumPy, SQL, NoSQL, ETL pipelines, dataset curation, feature engineering, sentiment analysis, Matplotlib, data visualization
Programming & Engineering Practice: Python, C++, SQL, Git/GitHub, unit testing, end-to-end testing (Playwright), code review, research-paper implementation

PROJECTS

Event-Driven RAG Agent for Document Ingestion & Q&A | [Code](#) | *Python, FastAPI, Inngest, LlamaIndex, Qdrant, OpenAI*
• Built an event-driven RAG system that asynchronously ingests and parses multi-page PDF documents and answers natural-language queries with grounded, citation-backed responses.
• Architected durable, fault-tolerant ingestion with Inngest (independently retryable parse → embed → store steps) over a semantic-retrieval layer using LlamaIndex chunking, OpenAI **text-embedding-3-large** embeddings, and a Qdrant vector store.
• Integrated gpt-4o-mini with context-aware prompt engineering for grounded answers over indexed documents; designed the pipeline to be modular and reconfigurable across chunking and embedding strategies.
• Exposed the workflow through a FastAPI backend with a Streamlit chat UI and Dockerized Qdrant — a decoupled, observable, end-to-end pipeline.

Multimodal Credit Risk Analysis using SEC XBRL & NLP | [Code](#) | *Python, Pandas, NumPy, Scikit-learn, NLTK/VADER, Git*
• Engineered an end-to-end data pipeline that parses ~41M raw SEC XBRL records (2022–2024, 12 quarters) into a curated multimodal dataset of ~35,000 companies with 47 features (34 financial + 13 NLP-derived).
• Trained and benchmarked four ML models (Random Forest, Gradient Boosting, Logistic Regression, SVM) for binary and multi-class corporate credit-rating prediction, reaching 97.9% (binary) and 93.8% (multi-class) accuracy with a Random Forest on financial features.
• Engineered 13 NLP features (sentiment, risk, safety, uncertainty, readability) from filing text using NLTK/VADER; validated results via per-feature-set evaluation (financial-only vs. combined), data-leakage checks, and logged model-comparison reports in modular, reproducible pipelines.

OPEN SOURCE CONTRIBUTIONS

Matplotlib | [PR #31422](#) merged into main (v3.11.0)
• Resolved issue #26620 by improving legend parameter documentation; reviewed and approved by core maintainer timhoffm and shipped in the v3.11.0 release of one of the most widely used Python visualization libraries.

JupyterLab | [PR #18610](#) merged | [PR #18668](#) approved
• Improved user-facing documentation and added Playwright end-to-end test coverage; both PRs cleared multi-round review with core maintainers, reflecting testing rigor and reproducibility.

CERTIFICATIONS & ACHIEVEMENTS

Google x Kaggle — 5-Day AI Agents Intensive: Agentic LLM workflows, LlamaIndex/LangChain orchestration, RAG system design.
GATE CSE: Ranked in the top 7.8% of ~210,000 candidates nationwide — secured M.Tech (AI/ML) at IIIT Naya Raipur.
Problem Solving: 200+ LeetCode problems solved; consistently top-ranked in programming coursework.